C.W

26.10.2026

(*) Find the condition that the following system of linear eqn:

$$x+y+z=6$$
$$x+2y+3z=10$$
$$x+2y+\lambda z=\mu$$

(i) has no soln (ii) a unique soln & (iii) more than one soln.

Solution:

The argumented matrix of the given system is: $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array}\right]$

Now, here we apply row echelon form, then we get,

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 0 & 3-\lambda & 10-\mu \\ 0 & 0 & \lambda-3 & \mu-10 \end{array}\right] \quad \begin{array}{l} R_2' = R_2 - R_3 \\ R_3' = R_3 - R_2 \end{array}$$

(ii) when, $(\lambda-3) \neq 0 \Rightarrow \lambda \neq 3$. then there will be one sol^{n}.

(iii) when $(\lambda-3) \neq 0 \Rightarrow \lambda = 3$ & $\mu - 10 = 0 \Rightarrow \mu = 10$ we get more than one sol^{n}

HW:

(*)

$x + y - z = 1$

$2x + 3y + \lambda z = 3$

$x + \lambda y + 3z = 2$

Find condition for

(i) has no sol^{n}.

(ii) a unique sol^{n}.

(iii) more than one sol^{n}.

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 0 & \lambda-3 & : & \mu-10 \\ 0 & 0 & \lambda-3 & : & \mu-10 \end{bmatrix} \begin{matrix} R_2' = R_3 - R_2 \\ R_3' = R_3 - R_2 \end{matrix}$$

Hence the required reduced system of linier eqn's are;

$$\left.\begin{matrix} x+y+z=6 \\ z(\lambda-3)=\mu-10 \\ z(\lambda-3)=\mu-10 \end{matrix}\right\} \quad \begin{matrix} x+y+z=6 \\ z(\lambda-3)=\mu-10 \end{matrix}$$

(i) when $\lambda-3=0 \Rightarrow \lambda=3$ and $\mu-10\neq 0 \Rightarrow \mu \neq 10$

then there has no sol^n.